

A1-brief

ISE-A1 —Problem and Stakeholder Evidence Brief

- Course: Intelligent Software Engineering | Platform: <https://learn.spaiq.ai>
- Release: 2026-09-21 16:30 | Deadline: 2026-09-28 16:30 (Asia/Shanghai) | Maximum: 5
- Author: Liwei Tang
- Every number below is regenerated by the scripts in `src/`; raw output is committed under `raw/measurements/`.

0. Scope statement

- **Workflow under study:** planning one semester of UCAS graduate courses at the 玉泉路 (Yuquanlu) campus, checking that the chosen combination has no timetable conflict, and deciding the order in which seats are taken inside the first-come-first-served selection window.
- **This brief claims:** producing a single input that can support both a conflict check and a grab-order ranking is impossible from any one source. It requires joining three heterogeneous sources by course code, and even after joining, a large share of courses still lacks one of the two field groups—including 6 of the 14 courses in the current plan whose semester is recorded nowhere. The repeat labour and the risk this creates justify a semester-sized intervention.
- **This brief does not claim:** that the selection system is defective; that automated tooling should replace the official submission procedure; or that the intervention covers graduation-audit rules.

1. Evidence ledger

Full provenance, hashes and dates: `docs/evidence-ledger.md`.

ID	Source type	What it establishes
E1	Personal operation record—two saved plan versions (2026-09-05, 2026-09-15) plus the planner backup	The plan is re-laid-out rather than patched: 22 cell positions dropped, 45 added, 0 edited in place

ID	Source type	What it establishes
E2	Official rule text —the selection instructions and the two official planning workbooks	Hard constraints: Yuquanlu only (course code position 18 = Y), no cross-campus selection, degree-course credit floor, first-come-first-served
E3	Official workbooks, structured import (4,877 records; 732 Yuquanlu records / 414 distinct courses)	0 of the 732 Yuquanlu records carry an 开课周 (week-range) value and 0 carry a 星期节次 (weekday/period) value ; the importer reports this defect four times
E4	Public course listing (2,991 entries; 347 Yuquanlu codes)	Carries the week range, the weekday/period, the capacity and the enrolled count for all 347 —but no campus field at all , so the Yuquanlu subset is only recoverable from the course-code rule
E5	Join of E3 and E4 by course code, plus the saved plan	Only 217 codes are in both; 197 exist without week/period/capacity; 130 exist without campus/semester. In the plan, 6 of 14 courses have no semester in any source —and 5 of those 6 are at $\geq 95\%$ of capacity

What would falsify the claim: a single source that already carries campus, semester, week range, weekday/period and remaining seats for the same courses. Searched on 2026-09-27*; record in raw/search-log.md (TODO: write up terms, date, results).

2. Stakeholder and workflow map

- **User:** the author —the person executing the whole procedure each semester.
- **Affected parties:** classmates at the same campus under the same rules; the supervisor, whose research time is affected by where courses land.
- **Decision makers:** the offering departments and the academic affairs office —they publish the plan and set the rules, so a rule change reopens the whole problem.
- **Constraints:** Yuquanlu campus only (code position 18 = Y); no cross-campus selection; a floor on degree-course credits; first-come-first-served within a fixed window; every source is a point-in-time snapshot.
- **Current responsibility:** entirely manual, with no tool support.

1. read the degree-programme requirements [nowhere in the data -- human input only]

2. assemble candidate sections + their times [three sources, joined by hand –
– E3, E4, E5]
3. collect capacity/enrolled per section [only the public listing has it –
– E4]
4. check for conflicts [BREAK POINT: three dimensions at once –
– E1, E3, E4]
5. decide the order to take seats in [second, different ordering problem –
– E5, priority-rules.md]
6. submit inside the selection window [first-come-first-
served; failure forces a re-plan -- E2]
7. re-plan whatever failed [returns to step 4 over the whole grid –
– E1]

Largest break point: steps 2-5 (E3 + E4 + E5). A clash needs same weekday **and** overlapping periods **and** overlapping week ranges; the two official workbooks carry neither the week nor the weekday/period, and the only source that carries them has no campus field, so building the input means joining sources that overlap only partially.

3. Problem / value statement

Who, where, how often. Each selection season the author must assemble roughly a dozen courses into one semester satisfying the programme's credit requirements, from a campus-restricted pool —414 distinct Yuquanlu courses recorded in the official workbooks, 347 in the public listing, and only 217 in both. The window between publication and closing is a few days and the plan is revised repeatedly inside it.

The pain, measured. Two independent things make this manual:

1. **The fields are split across sources.** The 4,877-record official import carries campus, semester, attribute and credits, but for its 732 Yuquanlu records it carries the week range **0 times** and the weekday/period **0 times** —the importer says so itself, four times. The public listing carries the week range, weekday/period, capacity and enrolled count for all 347 of its Yuquanlu entries, but it has no campus field, so those 347 can only be identified through the course-code rule, and it records no semester. Joining the two by course code leaves 197 codes with no week/period/seats and 130 with no campus/semester. Of the 14 courses in the current plan, **6 have no semester in any source, while 8 of the 13 that have a capacity figure are at or above 90% full** —and 5 of those 6 unknown-semester courses are at or above 95%.
2. **The check itself is three-dimensional.** A clash requires same weekday, overlapping periods and overlapping week ranges together. The measured week occupancy of the plan ranges from 5 to 19 teaching weeks depending on the course, and in the 2026-09-05 plan one slot (cell C5, Tuesday periods 1-4) hosts 新时代中国特色社会主义理论与实践 in weeks 11-18 and 自然辩证法概论 in weeks 2-5 and 7-10. A check using only weekday and period calls that a clash; only the week range shows it is not.

The workaround and its cost. Read the weekly worksheets across the semester by eye and re-check the whole grid whenever anything moves. Between the two saved versions 22 cell positions were dropped and 45 added with nothing edited in place: the plan is re-laid-out, and each re-layout forces a full re-check.

The intervention and its value measure. Assemble one frozen input by joining the sources on course code; detect three-dimensional clashes; and emit **two lists** —a conflict-free plan and a grab order ranked by deferability and seat scarcity (see `docs/priority-rules.md`). Value is measured on the same input with the same procedure: one full manual pass takes **[FILL: __ minutes, stopwatched, __ sections]** today and must fall below five minutes.

What is uncertain. (i) Whether the semester of the missing courses can be recovered at all, or whether it must be entered by hand once per semester —this shapes the whole design. (ii) How common the pain is beyond one account (E1 is one person's record plus one classmate's account). (iii) Whether the empty capacity cells in the official workbooks mean "no limit" or "not published" —currently treated as unknown, never as unlimited.

Why this is semester-sized and not a one-hour script. Four things carry the weight: joining three sources that overlap partially, and doing it reproducibly; getting the three-dimensional clash rule correct and testable; separating the two orderings (plan vs grab order) so the tool does not answer one question with the other; and making rejections legible enough that a human will trust the output.

4. Quantification

Measure	Value	How measured	Date
Baseline: one full manual conflict check	[FILL: __ min]	stopwatch over the [FILL: __] sections of the 2026-09-15 plan, 3 runs	[FILL]
Baseline: full re-checks per semester	≥ 2	two saved versions with a full re-layout between them (E1)	2026-09-25
Plan courses at $\geq 90\%$ of capacity	8 of 13	<code>src/rank_plan_priority.py</code> , seats from the frozen public listing	2026-09-27
Plan courses with no semester in any source	6 of 14	<code>src/rank_plan_priority.py</code> against the frozen official import	2026-09-27
Target value measure	< 5 min	same plan, same procedure, so the two numbers are comparable	—

Measure	Value	How measured	Date
Failure cost	clash (re-plan) / credit shortfall against the degree requirement / a seat lost inside the window	stated as consequences, not probabilities	—

Honesty note. No missed-conflict count is claimed: every revision so far was re-checked in full, so no clash is known to have occurred, and the cost is anchored on the repeat labour rather than on an error count.

5. Non-software alternatives

Option	How it works	Cost	Why it is insufficient (evidence)
A. Excel template + printed listings	fixed column headings, filled and compared by hand	[FILL]	the three-dimensional rule still lives in the operator's head; the sources change yearly and disagree in structure (E3, E4)
B. Existing planner / timetable tools	search for and try an existing tool	[FILL]	what was found targets institutional timetabling; none was found that joins these sources or carries the week range (search record: raw/search-log.md)
C. This proposal	frozen joined input -> three-dimensional clash detection -> plan + separate grab order	one semester of work	—

Why C wins on evidence rather than preference: neither A nor B can produce the joined input, and the measured C5 case shows the week range is what decides whether a shared weekday and period is a clash at all.

6. Scenarios

Normal. At the start of a season the author joins the sources to the Yuquanlu subset, checks conflicts, ranks the plan, and works down the grab order inside the window.

Boundary. A part-semester course sharing a slot with a full-semester course (the measured C5 case); two week ranges that touch but do not overlap (weeks 3 - 5 and 6 - 8 are **not** a

clash); a course at exactly capacity; a listing captured a week ago; a course sitting exactly on the code rule (position 18 = Y); and a course present in the plan that appears in no official workbook.

Abuse. The most obvious abuse is turning this into an **automatic course-grabbing program**: it would break first-come-first-served fairness and would likely violate the platform's selection rules, so it is explicitly excluded. Others: reading another student's enrolment records; fabricating section data to make a plan look feasible; presenting an unfrozen snapshot as current; and treating an empty capacity cell as "unlimited seats" rather than "unknown" .

7. Semester scope

- **In scope:** join the sources on course code into one frozen input; detect weekday x period x week-range clashes; rank candidates by the stated priority rules; emit a plan **and** a separate grab order; explain each rejection by naming the overlapping week range.
- **Non-goals:** no automatic submission or any write access to the selection system; no graduation-audit rules; no scraping of enrolment lists; no attempt to infer the degree-programme requirement automatically.
- **Assumptions:** the sources stay obtainable so a frozen snapshot with a hash is possible; week ranges are integer intervals, possibly with gaps; the must-take set is supplied by a human and is not derived.
- **Stop conditions:** if a course's semester cannot be recovered from any source and cannot be supplied by hand, stop and report it rather than assuming the course is postponable. If the public listing ever stops publishing the week/period fields, narrow the scope to a timetable-consistency checker (still three-dimensional) rather than substituting guessed data.

8. AI-use and source-verification log

See `ai/ai-use-log.md`. One intermediate conclusion was **refuted by a source** during this work and the wording of this brief was changed as a result: an earlier draft claimed that no machine-readable source carries the week range or the weekday/period, which the public listing disproved. The two priority dimensions in section 3 come from the author, not from a model.

9. Repository and submission

- Repository: <FILL: repo URL>
- Commit / tag: a1-v1
- Platform summary: `docs/submission-summary.md` (verified with `python3 src/count_words.py`)
- Priority model: `docs/priority-rules.md`